RISK ASSESSMENT PRACTICE

Technical Report & Documentation

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Program: Red Teaming Internship

Week: Week 2

1. Overview

This report documents Activity 6 of Week 2 of the CyArt Red Teaming Internship: Risk Assessment Practice. The objective was to perform a quantitative risk assessment using the ALE (Annualized Loss Expectancy) formula for a ransomware scenario, document the calculations in Google Sheets, and build a 5x5 risk matrix to visually score the risk by Likelihood and Impact. These are foundational skills in cybersecurity risk management used by risk analysts, GRC professionals, and red teams to communicate the business impact of threats.

2. Tool Used

- Google Sheets — for performing ALE calculations and building the risk matrix

3. Key Risk Terminology

Term	Full Form	Definition
SLE	Single Loss Expectancy	The monetary loss expected from a single occurrence of a specific threat event.
ARO	Annualized Rate of Occurrence	The estimated frequency (per year) that a threat is expected to occur.
ALE	Annualized Loss Expectancy	The total expected annual loss from a threat: ALE = SLE x ARO.
Risk	Risk Score	A measure of likelihood and impact used to prioritize threats in a risk matrix.

4. ALE Calculation — Ransomware Scenario

4.1 Scenario Description

The organization faces a ransomware threat. Based on historical data and threat intelligence, the following values were established for the risk calculation:

Ransomware Risk Scenario — Input Values

Threat: Ransomware attack encrypting all corporate file servers

SLE (Single Loss Expectancy): $10,000

ARO (Annualized Rate of Occurrence): 0.2 (i.e., once every 5 years)

Formula: ALE = SLE x ARO

4.2 ALE Calculation

Step	Formula / Value	Result
Step 1 — Identify SLE	`SLE = $10,000`	**$10,000.00**
Step 2 — Identify ARO	`ARO = 0.2`	0.2 (once every 5 years)
Step 3 — Apply Formula	`ALE = SLE x ARO`	$10,000 x 0.2
Final Result — ALE	`ALE = $10,000 x 0.2`	**ALE = $2,000 per year**

Interpretation: The organization can expect to lose $2,000 per year on average due to ransomware attacks, based on the current threat rate and loss estimate. This figure is used to justify the cost of security controls — any control costing less than $2,000 per year delivers a positive return on security investment (ROSI).

4.3 Google Sheets Implementation

The ALE calculation was implemented in Google Sheets using cell formulas. The spreadsheet was structured as follows:

Cell	Label	Formula / Value Entered	Output
B2	SLE	`10000`	**$10,000.00**
B3	ARO	`0.2`	**0.2**
B4	ALE	`=B2*B3`	**$2,000.00**
B5	Risk Level	`=IF(B4>5000,"HIGH",IF(B4>1000,"MEDIUM","LOW"))`	**MEDIUM**
B6	Recommended Control Budget	`=B4*0.8`	**$1,600.00**

5. Risk Matrix — 5x5 Likelihood vs. Impact

5.1 Risk Matrix Scoring Scale

A 5x5 risk matrix was used to visually plot risks by scoring both Likelihood (how often the threat occurs) and Impact (how severe the consequences are) on a scale of 1 to 5. The Risk Score is calculated as: Risk Score = Likelihood x Impact.

Score	Likelihood	Impact	Risk Score Range	Risk Level
1	Rare (< once in 10 yrs)	Negligible	**1 – 4**	**LOW**
2	Unlikely (once in 5–10 yrs)	Minor	**5 – 9**	**LOW-MED**
3	Possible (once in 2–5 yrs)	Moderate	**10 – 14**	**MEDIUM**
4	Likely (once per year)	Major	**15 – 19**	**HIGH**
5	Almost Certain (> 1/yr)	Catastrophic	**20 – 25**	**CRITICAL**

5.2 The 5x5 Risk Matrix

The matrix below plots Likelihood (rows, 1–5) against Impact (columns, 1–5). Each cell shows the Risk Score (Likelihood x Impact) and its colour-coded risk level. The ransomware scenario is marked with a star (★).

Likelihood \ Impact	1 Negligible	2 Minor	3 Moderate	4 Major	5 Catastrophic
5 Almost Certain	5	10	15	20	25
4 Likely	4	8	12	16	20
3 Possible	3	6	9	12	15
2 Unlikely	2	4	6	8 ★ Ransomware	10
1 Rare	1	2	3	4	5

LOW (1–4)	LOW-MED (5–9)	MEDIUM (10–14)	HIGH (15–19)	CRITICAL (20–25)

5.3 Ransomware Scenario Score

Parameter	Value	Justification
Likelihood Score	**2 — Unlikely**	ARO = 0.2 means once every 5 years
Impact Score	**4 — Major**	Data encryption, downtime, ransom payment, reputational damage
Risk Score	**2 x 4 = 8**	Likelihood x Impact
Risk Level	**LOW-MEDIUM**	Score 8 falls in the 5–9 range
Priority	Monitor and treat	Should be addressed with controls given financial impact

6. Risk Register Entry

Based on the ALE calculation and risk matrix scoring, the ransomware scenario was entered into the organization's risk register as follows:

Risk	Threat	SLE	ARO	ALE	Score	Recommended Control
Ransomware Attack	Malicious email /	$10,000	0.2	**$2,000**	**8 / 25**	Email filtering, offline backups,

Risk	Threat	SLE	ARO	ALE	Score	Recommended Control
	unpatched system					endpoint protection, staff training
[Add from lab]						

7. Key Observations and Findings

- The ALE formula (ALE = SLE x ARO) provides a simple but powerful quantitative method to express risk in financial terms that non-technical stakeholders can understand.
- An ALE of $2,000 per year means any security control costing less than $2,000 annually represents a positive return on security investment.
- The 5x5 risk matrix adds a qualitative dimension by scoring both how likely a threat is and how severe its consequences would be.
- Ransomware scored 8 out of 25 on the risk matrix — Low-Medium — reflecting its relatively low occurrence rate (ARO 0.2) despite its high impact.
- If the ARO were increased to 1.0 (annual ransomware attack), the ALE would rise to $10,000 and the risk score would jump to 20 (Critical), dramatically changing the priority.

8. Challenges Faced

- Estimating realistic ARO values requires access to industry threat data or historical incident records, which may not be available in a lab environment.
- The SLE value can vary significantly depending on what losses are included (direct costs, downtime, regulatory fines, reputational damage), making standardization important.
- Building a colour-coded matrix in Google Sheets required applying conditional formatting rules across all 25 cells to reflect the correct risk tier.

9. Learnings and Takeaways

- Quantitative risk assessment using ALE translates technical threats into business language, enabling informed decisions about security spending.
- Risk matrices are a fast and visual way to communicate threat priorities to management and non-technical teams.
- Combining ALE (quantitative) with a risk matrix (qualitative) provides a more complete picture than either method alone.
- Google Sheets is an effective tool for lightweight risk analysis and can be shared and updated collaboratively across teams.

10. Next Steps

- Expand the risk register to include additional threat scenarios such as insider threats, DDoS attacks, and data breaches.
- Apply sensitivity analysis to the ALE calculation by adjusting SLE and ARO values to model best-case and worst-case scenarios.
- Explore dedicated GRC (Governance, Risk, and Compliance) tools such as OpenRMF or risk management modules in security platforms.
- Present the risk register to a mock stakeholder audience to practice communicating findings in business terms.